

JESSE CAMPBELL

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Education

University of Illinois at Chicago
M.S. Mathematical Computer Science

Aug. 2025 – May 2026
GPA: 4.00/4.00

Duke University/Duke Kunshan University (Dual-Degree)
B.Sc. Applied Mathematics and Computational Science

Aug. 2021 – May 2025
Summa Cum Laude, GPA: 3.93/4.00 (Top 5%)

Technical Skills

Research Specialties	Machine learning theory, Differential privacy, Algorithms
Programming Languages	Python, Java, C, SQL, Mathematica, Julia, L ^A T _E X ROOT
Languages	English (Native), Mandarin (Advanced)

Peer-Reviewed Publications

1. Jesse Campbell and Chunjiang Zhu. Differentially private counting queries on approximate shortest paths. In *Combinatorial Optimization and Applications: 17th International Conference, COCOA 2024*, Beijing, China, Dec. 2024., Lecture Notes in Computer Science, vol 15434. Springer.

Professional Experience

Teaching Assistant University of Illinois, Chicago

August 2025 – Present

- Leading discussion sections and grading assignments for undergraduate mathematics courses.

Software Testing Team Intern KONE Elevators China

May 2023 – August 2023

- Diagnosed the internal API and integrated with database coded in SQL by personally writing over 100 automation scripts in Python using modules such as Selenium to simulate browser actions.
- Completed internship totally in Mandarin, Chinese.

Research Projects

On the Contradiction Graph of a Hypothesis Class

November. 2025 – Present

- Project presenting several new results regarding the contradiction graph of a hypothesis class.

Chromatic Symmetric Functions

Jan. 2023 – Apr. 2025

- Completed 40-page undergraduate thesis studying a 30-year open problem in algebraic combinatorics. Presented work at thesis exhibition conference.

Graph Counting Queries with Differential Privacy

May 2024 – Aug. 2024

- Completed a project during the GraLNA NSF REU at the University of North Carolina working on release of counting queries and shortest path distances in graphs under differential privacy.

GNNs for dRICH Classification

May 2022 – Aug. 2022

- Research assistant for High energy physics 101 (HEP 101) cohort. Designed message passing graph neural network for classification of particles based on simulation dual ring-imaging Cherenkov detector (dRICH) data.

Honors and Awards

CRA NSF REU Travel Award Awarded travel grant for \$1200 from the Computing Research Association in Washington D.C. to showcase research at COCOA 2024 conference in Beijing, China.

Nov. 2024

Seminars & Teaching

Teaching Assistant

- Calculus II (UIC, Spring 2026)
- Finite Math for Business (UIC, Fall 2025)
- Discrete Math for Computer Science (DKU, Spring 2024 and Spring 2025)
- Probability and Statistics (DKU, Spring 2025)

UIC CS Theory Seminar Presenter

Aug. 2025 – Present

UIC Graduate Combinatorics, Algorithms and Probability Seminar Presenter

Aug. 2025 – Present

DKU Discrete Mathematics Seminar Co-organizer and presenter

Jan. 2023 – May 2025

Extracurricular Experience

Math Circles of Chicago Volunteer

Aug. 2025 - Present

DKU Jazz Club Founder and president

Aug. 2022 – May 2025